

Analysis of the effluents of oxidation reactors in terephthalic acid production

Abstract

The production of purified terephthalic acid (PTA) is performed by direct oxidation of p-xylene using air as oxidant and acetic acid as solvent. The reaction takes place at about $195\pm 5^{\circ}\text{C}$ and it is catalyzed by Co, Mn, and Br. The crude product is then purified by hydrogenation in water at about 280°C . Notwithstanding the huge capacity of world scale plants, such as 700 kton/year, the specifications on PTA are really stringent, being the purity required 99.97 %. The crude product purity affects most of the quality parameters of PTA, indeed most of the side reactions take place in the oxidation section, and lead to byproducts poorly soluble in both acetic acid and water. In this report we determined how quality parameters of the effluent of oxidation reactors depend on their composition, identifying a compound, or class of compounds, related to each of the main quality parameters. Both main and secondary-oxidation reactors were considered, showing the key role of the secondary oxidation reactor in determining the quality of terephthalic acid.





TABLE OF CONTENTS

INTRODUCTION	4
EXPERIMENTAL SECTION.	5
RESULTS AND DISCUSSION	5
Reactors Effluent.....	6
Transmittance % at 340 nm.	6
b* value.....	8
Metal content	10
Effect of the post oxidizer on the product quality.....	12
APPENDIX 1.....	18
APPENDIX 2.....	29
REFERENCES	46

TABLE OF FIGURES

- Figure 1.** Transmittance % at 340 nm measured on the solid product present in the effluent of the oxidation reactors
- Figure 2.** TR of reaction's intermediate as a function of the concentration. Each concentration range has been chosen in accordance with the quantities actually detected in the product.
- Figure 3.** TR of reaction's intermediate as a function of the concentration. Each concentration range has been chosen in accordance with the quantities actually detected in the product. 1,2,4-Benzene tricarboxylic acid $\diamond$; 1,2,3-Benzene tricarboxylic acid $\square$; Benzoic acid $\odot$; Benzophenone-4,4'-Dicarboxylic acid $\circ$; 9,10-Dihydro-9,10-dioxo-2,3-Anthracene dicarboxylic acid $\blacklozenge$; Anthracene-9,10-Dicarboxylic acid $\blacktriangle$
- Figure 4.** b* value of TA as a function of its TR
- Figure 5.** Effect of 's by-products on the b* value of the product
- Figure 6.** b* value as a function of the concentration of 9,10-dihydro-9,10-dioxo-2,3-anthracene dicarboxylic acid
- Figure 7.** Concentration of metals (Mn + Co + Fe) as a function of the concentration of 1,2,4- Benzene tricarboxylic acid in TA
- Figure 8.** Concentration of metals Mn as a function of the concentration of 1,2,4- Benzene tricarboxylic acid in TA
- Figure 9.** Total 4-CBA reduction percentage as a function of the total 4-CBA inlet mass
- Figure 10.** Total p-toluic reduction percentage as a function of the total p-toluic inlet mass
- Figure 11.** By products percentage variation as a function of the total by products inlet mass.
- Figure 12.** Reduction of 4-CBA concentration in solid as a function of the 4-CBA concentration in the inlet solid the inlet solid
- Figure 14.** Variation of byproducts concentration in solid as a function of the byproduct's concentration in the inlet solid
- Figure 15.** Variation of TR as a function of TR of the incoming solid. The variation has been calculated considering the average of the two reactor products weighed on the production rate of each reactor.
- Figure 16.** Variation of solid's b* value as a function of b* value of the incoming solid. The Variation has been calculated considering the average of the two reactor products weighed on the production rate of each reactor.
- Figure 17.** Plant scheme.



INTRODUCTION

The production of purified terephthalic acid (PTA) is performed by direct oxidation of p-xylene using molecular oxygen – which main source is air - as oxidant and acetic acid as solvent. The reaction takes place at about $195\pm 5^{\circ}\text{C}$ and it is catalyzed by Co, Mn, and Br. The crude product is then purified by hydrogenation in water at about 280°C . Notwithstanding the huge capacity of world scale plants, such as 700 kton/year, the specifications on PTA are stringent, being the purity required 99.97 %. Typical PTA specifications are here reported as reference:

Description	Unit	Expected	Guarantee
Acid value	mg KOH / g	675 ± 2	675 ± 2
Moisture	wt. %	0.05	0.1 max.
Ash content	ppm (wt.)	< 5	6 max.
Iron (Fe)	ppm (wt.)	< 0.5	2 max.
Total heavy metals (3)	ppm (wt.)	< 2	3 max.
4-Carboxybenzaldehyde	ppm (wt.)	10	25 max.
p-Toluic acid	ppm (wt.)	60	150 max.
APHA color			
(5% DMF solution)	-	< 5	10 max.
Transmittance@340 nm	%	90	n.a.
b*	-	0.8	1.5 max.
Delta Y		1.2	n.a.
Particle size distribution			
$\geq 250 \mu\text{m}$	wt %	3% max.	5% max.
$\leq 40 \mu\text{m}$	wt %	20% max.	20% max.
Average Particle size			
(wet analysis)	μm	120 ± 20	n.a.

During purification the crude product is solubilized in high temperature water, hydrogenated, then crystallized and dried. As a consequence, moisture and particle size distribution are affected mainly by the purification conditions. The intermediates' content (4-carboxybenzaldehyde and p-toluic acid) in PTA depends both on the purification efficiency and the intermediates' content of the crude product fed to purification. The crude product purity affects also metals content and optical properties (APHA color, transmittance @340 nm, b* value). ΔY , which depends on the carbon particles released by the hydrogenation catalyst, is not affected by the



impurities contained in the CTA. Most of the side reactions take place in the oxidation section, and lead to byproducts poorly soluble in both acetic acid and water. These compounds are generally considered responsible for the yellow color (i.e. the b^* value) and metal content of terephthalic acid. In the last decades a great effort has been devoted to the identification of by-products playing a role in the optical properties and metal content in PTA/CTA and also to determine the best process conditions to reduce their formation [1-11]. Even if some general correlations have been found, neither the compounds representing an issue during polymerization are yet clearly identified, nor specific relationships between by-products and PTA quality parameters have been defined. The aim of this report is to determine how quality parameters of the effluent of oxidation reactors depend on their composition, identifying a compound, or class of compounds, related to each of the main quality parameters.

Both main and secondary-oxidation reactors were considered, in order to gather also a better understanding of the specific effect of the secondary oxidation reactor.

EXPERIMENTAL SECTION.

The tests were performed in the PTA plant located in Ottana. The plant has two reactors operated in parallel and the combined effluent is sent to a single post-oxidation reactor.

The slurry coming from the two oxidation reactors (R1301A, R1301B) was sampled separately. The effluent of the single post oxidation reactor, fed from the two reactors, was sampled one residence time later, within 30 and 40 minutes, as a function of the operating conditions. Each sample was filtered under vacuum, and the mother liquor was separated for the analysis. The solid was then dispersed into acetic acid 96% at 70 °C keeping a solid/liquid ratio of 1:2. The dispersion was agitated for 10 minutes and then filtered and washed on the filter with acetic acid 96% at 70 °C (solid/liquid ratio 1:1). The solid was dried up at ambient temperature, under hood, for 24 hours and then analyzed. This procedure was optimized in order to remove as much as possible the mother liquor from the solid, minimizing contamination. The complete analytical results are reported in Appendix 1. For each test, a material balance has been calculated based on analytical results, plant data and, when available, online analyzers data. Material balances are reported in Appendix 2.

RESULTS AND DISCUSSION

The main parameters considered were the optical properties: Transmittance % at 340 nm, b^* value and metal content in the solid product. Optical properties are important indexes of the product quality while the metal content in the solid represents an environmental concern being, after the purification process, sent outside battery limits together with wastewater. Moreover, a significant amount of metals in the solid product results in an appreciable loss of catalyst. For sake of clarity each parameter will be considered separately. The findings relevant to solid sampled from the reactors (hereinafter TA) are discussed in the first section and then the effect of the post oxidizer on the product quality will be discussed in second section.



Reactors Effluent

Transmittance % at 340 nm.

In PTA manufacturing transmittance % at 340 nm (hereinafter TR) is defined as the transmitted percentage of a 340 nm radiation incident on an alkaline solution of terephthalic acid. TR is generally considered as an indicator of the purity of the product, and indeed it shows a proportional trend as a function of the byproducts contained in TA, see Figure 1.

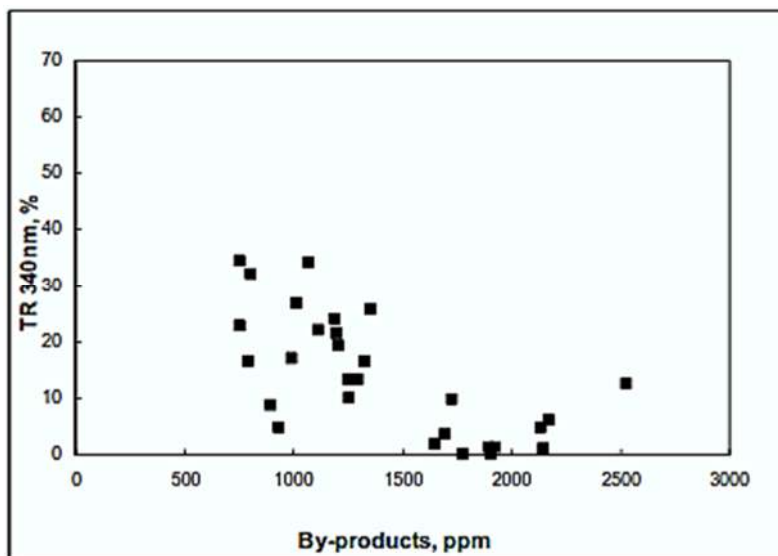


Figure 1. Transmittance % at 340 nm measured on the solid product present in the effluent of the oxidation reactors.

In order to clarify the effect of each byproduct and also any eventual contribution from the reaction intermediate, TR of each of the determined by-product and intermediate was measured as a function of the concentration. The concentration range, for each compound, was chosen according to the concentrations actually measured in the product. The results are reported in Figure 2 and Figure 3.

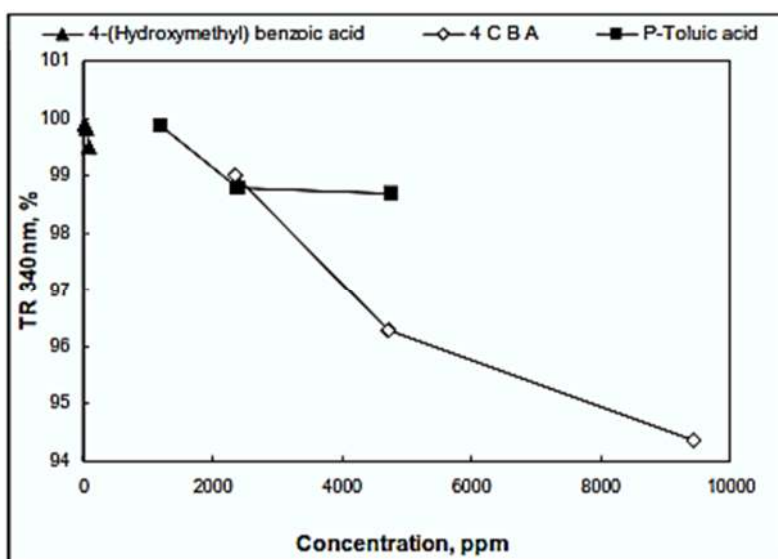


Figure 2. TR of reaction's intermediate as a function of the concentration. Each concentration range has been chosen in accordance with the quantities detected in the product.



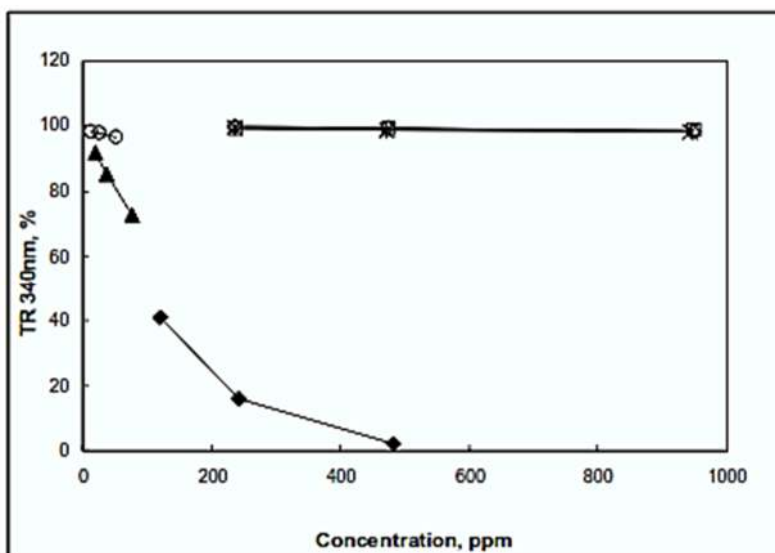
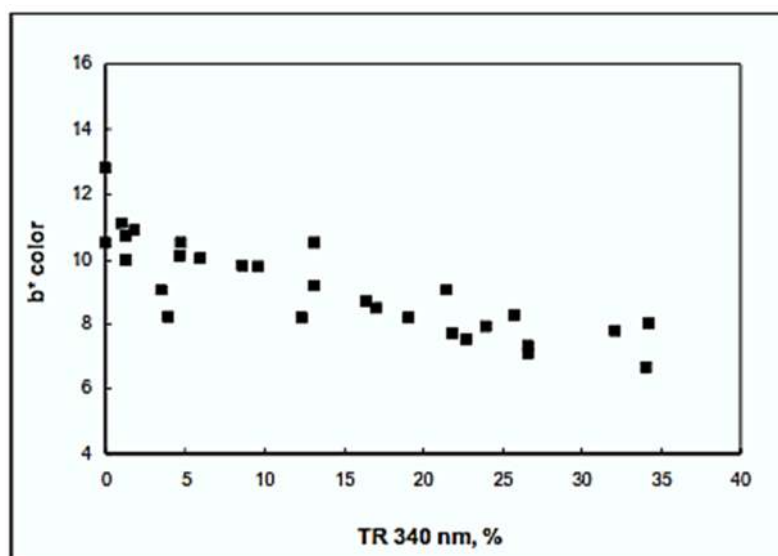


Figure 3. TR of reactions intermediates as a function of concentration. Each concentration range has been chosen in accordance with the quantities detected in the product. 1,2,4- Benzene tricarboxylic acid ◇; 1,2,3- Benzene tricarboxylic acid □; Benzoic acid ○; Benzophenone-4,4- Dicarboxylic acid ◊; 9,10-Dihydro-9,10-dioxo-2,3-Anthracene dicarboxylic acid ◆; Anthracene-9,10- Dicarboxylic acid ▲

As can be seen, reaction intermediates have a negligible effect on the TR. The more effective is 4-carboxy benzaldehyde (4-CBA) that in concentration of 1% reduces TR of about 6% (100% being a 2M solution of KOH). Similar behavior is shown by the by-products with a single benzene ring such as tri-carboxylic acid. On the contrary, benzophenone-type and anthracene-based compounds seem to be more effective in respect of TR, few hundreds ppm of these compounds can drop the TR down to 0 %. This effect is consistent with the fact that some of the by-products contain conjugated double bonds and then absorbing in the UV region. The formation of these compounds is favored in oxygen starving reactive mixtures and also when an excess of catalyst is present.

b value*

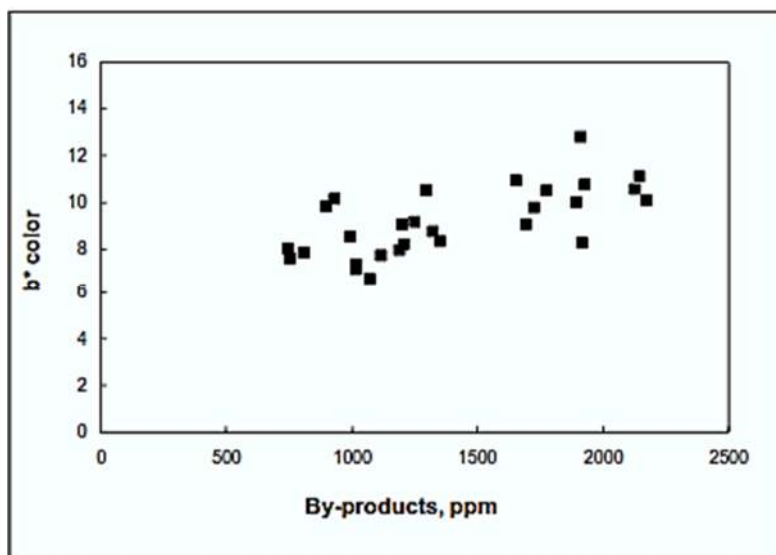


An additional optical property generally checked is the color through the CIE Lab* System colorimetric **Figure 4.** *b** value of TA as a function of its TR



coordinates where a^* stands for red-green (positive-negative values), L for light-dark (positive-negative values), and b^* for yellow-blue (positive-negative values) with reference to a standard. In PTA manufacturing the most commonly used is b^* value, that is a specification parameter for PTA, and, such as TR, is generally linked to the presence of impurities. TR and b^* value for TA as can be seen in Figure 4, show a proportional trend. This trend suggests that b^* value too should be dependent on the concentration of reaction by-products in TA, Figure 5.

Figure 5. Effect of 's by-products on the b^* value of the product.



The TR is measured in the UV region of the electromagnetic spectrum, and, as it has been demonstrated, is influenced by by-products absorbing in this region. b^* value is an index of the color in the visible region, then can be influenced only by colored compounds – i.e. compound with significant absorption in the visible region. The UV-Vis spectrum of some reaction intermediates and by-products solution has been collected in order to better understand which of them can increase the b^* value and it has been found that the main players are again compounds with more than one benzene ring bridged through double bonds, forming a conjugated double bond pattern. These compounds have, besides the strong absorption in the UV region, a long tail in the visible region of the electromagnetic spectrum. In particular, the main effect seems to be due to 9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic. The trend of b^* value as a function of the concentration of 9,10-dihydro-9,10-dioxo-2,3-anthracene dicarboxylic acid is reported in Figure 6.



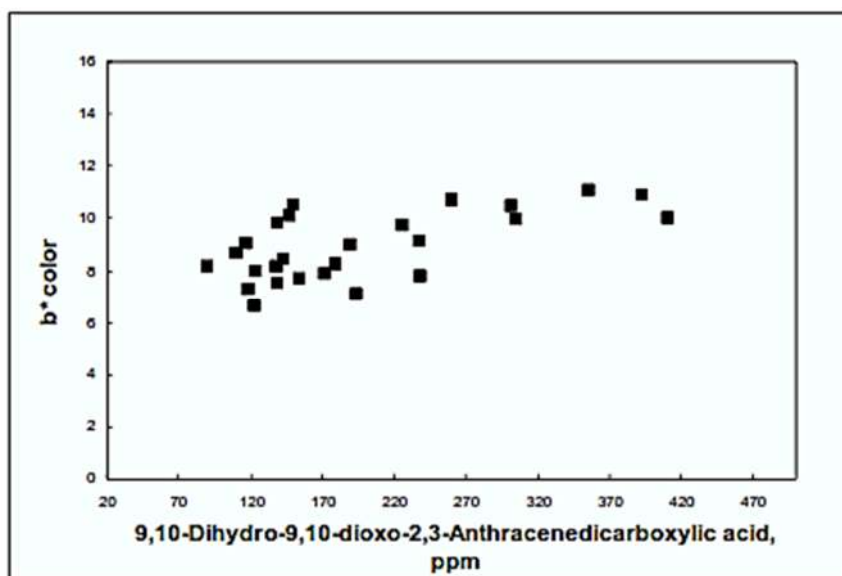


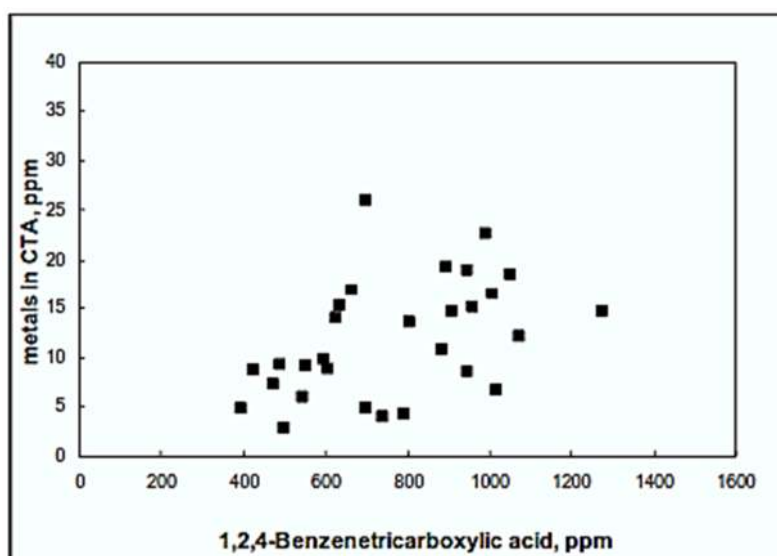
Figure 6. b^* value as a function of the concentration of 9,10-dihydro-9,10-dioxo-2,3-anthracene dicarboxylic acid.

Metal content

Metal content in Crude Terephthalic Acid (CTA) is one of the main issues in the production of PTA.

Indeed, all the metal catalysts present in CTA are sent to the purification section and then lost, increasing the production costs. Moreover, the presence of metals in the purification section can increase the corrosive capability of the streams. Finally, the metals entering into the purification section will flow outside battery limits, generally to biological treatment, causing environmental problems or requiring a pretreatment of wastewater. As it has been reported by Partenheimer [12] the metals present in CTA can be hydrogen bonded to some of the by-products, in particular tri-carboxylic acids. This hypothesis has been checked, and we found that a proportional trend does exist between the concentration of metals (sum of Mn, Co and Fe) and the concentration of 1,2,4-Benzenetricarboxylic acid in TA, reported in Figure 7.

Figure 7. Concentration of metals (Mn + Co + Fe) as a function of the concen-



tration of 1,2,4-Benzene tricarboxylic acid in TA



Keeping in mind that 1,2,4-benzenetricarboxylic acid has no effect on TR, these findings are also in agreement with the plant experience showing that the concentration of metals in TA is not correlated to TR, and that metals can be hardly removed increasing the product washing on the filter cloth. As a matter of fact, the major effect of washing on the filter cloth is changing clean acetic acid for mother liquor, whereas the effectiveness in breaking hydrogen bonds is very poor. Moreover, it appears that 1,2,4-benzenetricarboxylic acid has a particular affinity for manganese as evidenced in Figure 8, where the concentration of Mn is plotted as a function of the concentration of the acid and a clear proportional trend appears.

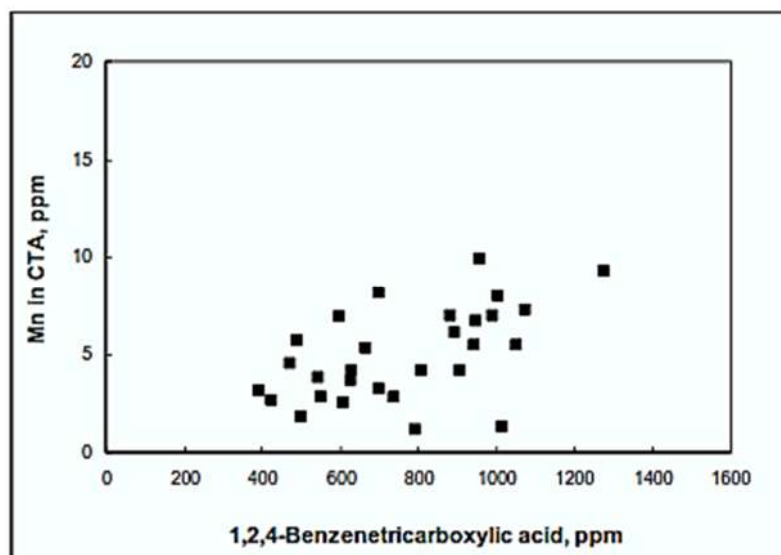


Figure 8. Concentration of metals Mn as a function of the concentration of 1,2,4-Benzene tricarboxylic acid in TA

Effect of the post oxidizer on the product quality

In this section the effects of the post oxidizer on the product quality is analyzed. In most of the commercialized terephthalic acid technologies the effluent of the reactor is sent to the first crystallizer where some air is introduced to perform a further oxidation step. The aim of this device is to reduce the amount of reaction intermediates in the stream sent to filtration section, enhancing the product quality. It is evident how a good knowledge of the effect of this apparatus on product quality is important during the design phase and in the material balance development. Reaction intermediate and by-products are present in the reactor effluent both in solution and co-crystallized. Soluble impurities will be largely recycled back to the oxidation reactor, while co-crystallized ones will affect the quality of the final product. Provided that crystallization occurs by flash, over saturation areas can be created during the crystallization, causing the co-crystallization of compounds even if they have a very large solubility difference with terephthalic acid. One-benzene-ring by-products have molecular dimensions similar to terephthalic acid, and then they can easily occupy a site in terephthalic acid crystalline lattice. Moreover, also bigger compounds can be included in terephthalic acid lattice as interstitial inclusions. Considering the total mass of each by-product/intermediate - the sum of solubilized and co-crystallized masses - its variation across the post-oxidizer represents the efficiency as purifying reactor. The post oxidizer reduces the



total mass of 4-CBA and p-toluic acid by about 60% and 80% respectively, without any dependence from the inlet mass, as shown in Figure 9 and 10.

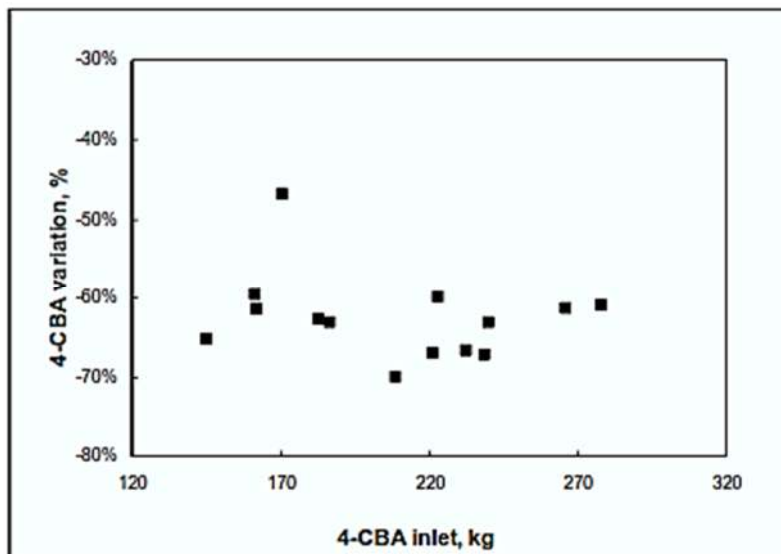


Figure 9. Total 4-CBA reduction percentage as a function of the total 4-CBA inlet mass.

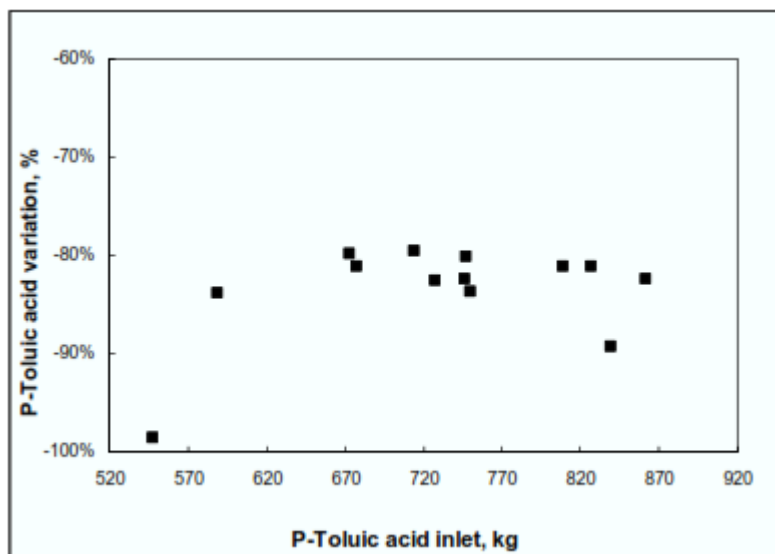


Figure 10. Total p-toluic reduction percentage as a function of the total p-toluic inlet mass.

In Figure 11, the effect of the post oxidizer on the by-products in the incoming stream is showed.

Quite surprisingly, it appears that also the by-products are reduced by further oxidation, and, apparently, this reduction depends from the total amount present in the inlet stream. Indeed, as the inlet mass increases, the effect of the post oxidizer seems to decrease moving from about -30% for inlet mass lower than 2000 kg to -10% for inlet mass higher than 4000 kg.

The observation of the effect of post oxidizer on the total amount of intermediates and by-products, indicates that this unit operation has a real important role in refining the effluent of the reactor, with reference to both intermediates and by-products. It shall be considered that, while the effect on the total amount of impurities



depends on the action of the equipment as oxidizer, the effect on the intermediate and by-products present in the solid is related to both the effect as oxidizer and the effect as crystallizer.

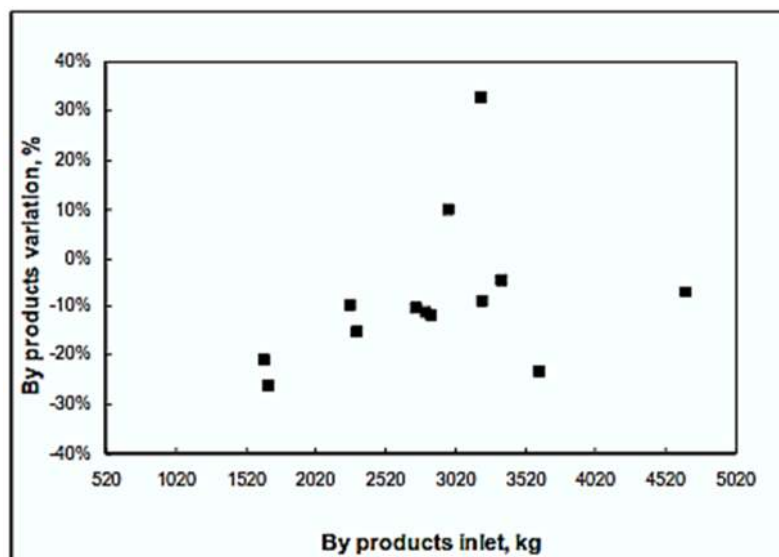


Figure 11. By products percentage variation as a function of the total by products inlet mass.

As a matter of fact, besides crystallization through nucleation and growth of the crystals, digestion takes place. This phenomenon includes solubilization of smaller particles and their re-crystallization on bigger ones, and a fast mass exchange between particles' surface and mother liquor. The first mechanism enhances the growth of bigger particles at the expense of small ones, the second one reduces crystalline defects, such as impurities inclusion in the crystal. In order to have some hints on the solid product quality enhancement due to the post oxidizer, the effect on intermediate and by-products present in the solid has been considered. In Figure 12 the % reduction of 4-CBA as a function of the concentration present in the solid coming in the post oxidizer is shown.

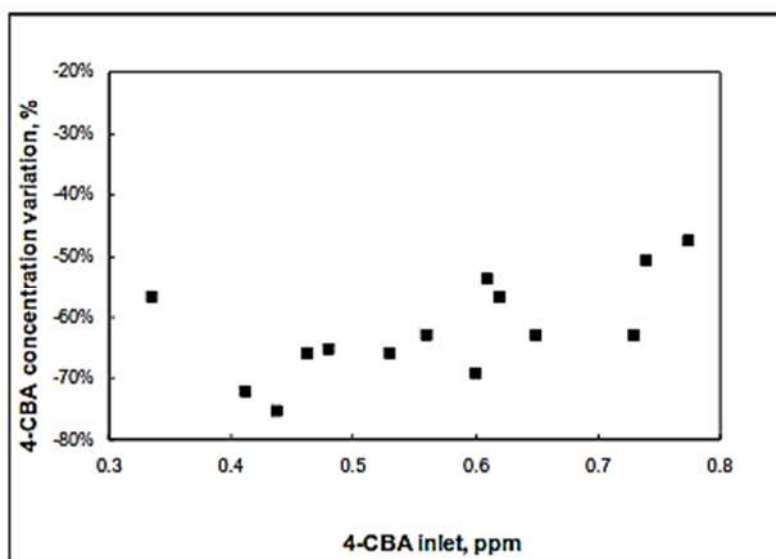


Figure 12. Reduction of 4-CBA concentration in solid as a function of the 4-CBA concentration in the inlet solid.



As can be observed, the reduction changes from -80 % to -50% as the concentration in the incoming solid increases. The variation of p-toluic acid present in the solid seems to have a weaker dependence on the inlet concentration, Figure 13, the reduction being almost constant at 85%.

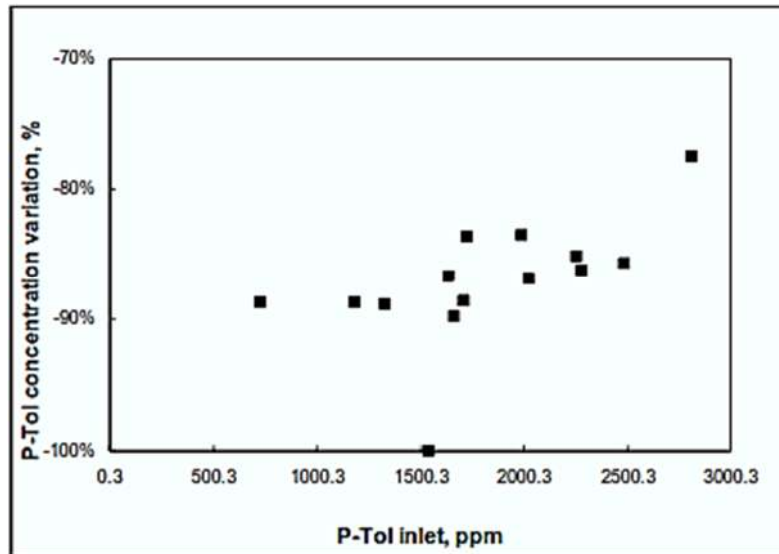


Figure 13. Reduction of p-toluic acid concentration in solid as a function of the p-toluic acid concentration in the inlet solid.

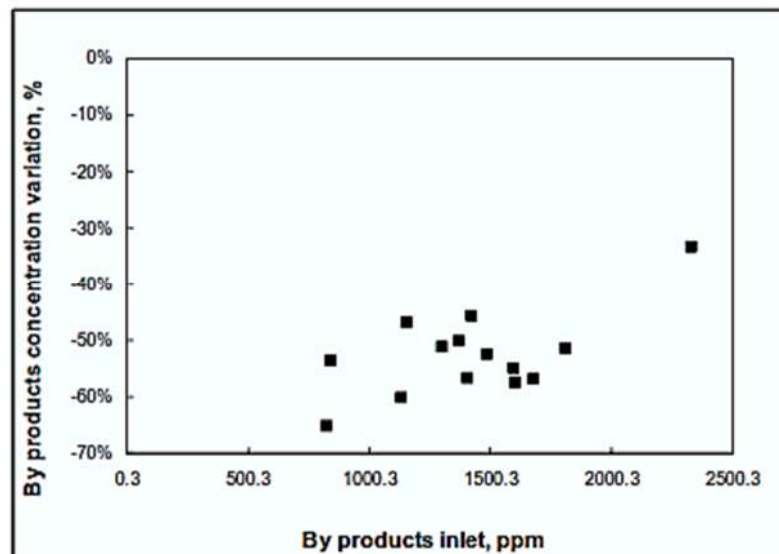


Figure 14. Variation of byproducts concentration in solid as a function of the byproduct's concentration in the inlet solid.

Reduction of by-products present in the solid is around 50%, apparently without dependence on the concentration in the inlet solid, Figure 14.

Considering the optical properties, Figure 15 and 16, the average increase in TR is 300% independently of the TR of the incoming solid, while the b^* value variation shows an increasing trend going from -20% to about 0% as the value of the incoming solid goes from 8 to 10.



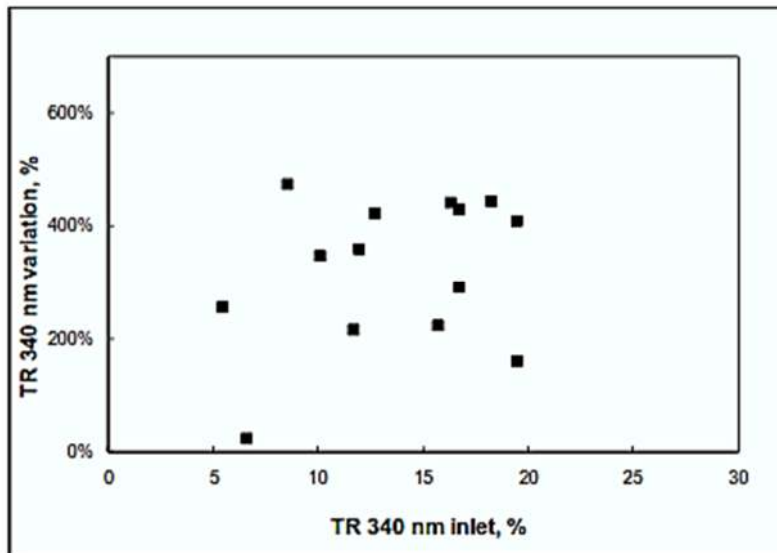


Figure 15. Variation of TR as a function of TR of the incoming solid. The variation has been calculated taking into account the average of the two reactor products weighed on the production rate of each reactor.

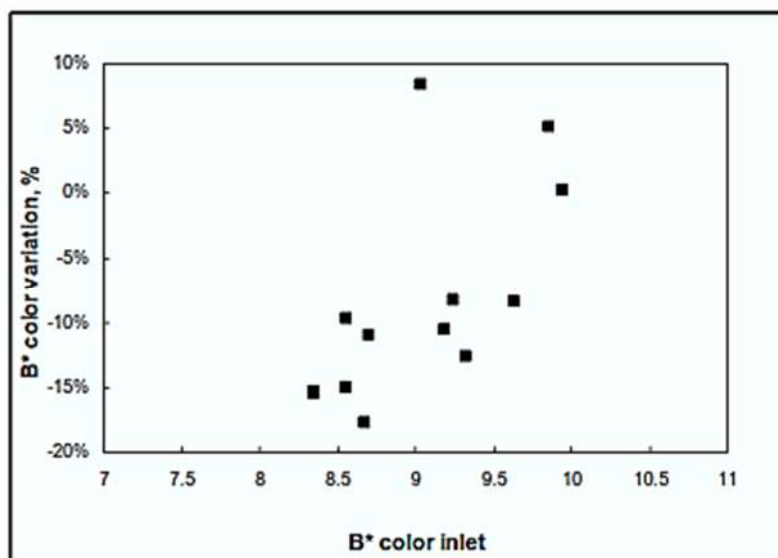


Figure 16. Variation of solid's b^* value as a function of b^* value of the incoming solid. The variation has been calculated taking into account the average of the two reactors' products weighed on the production rate of each reactor.

Some last considerations arise comparing averaged data for main quality parameters relevant to both post oxidizer outlet and dried CTA, Table II.



Table II. Quality data for the solid product coming out from the post oxidizer and for the dried CTA.
If not indicated otherwise, CTA data are averaged considering the value of the sampling days.

	Post Oxidizer			CTA			Notes
	min	average	max	min	Average	max	
TR 340 nm	8.4	28.7	39.2	19.9	30.4	40	
4-CBA	0.2	0.3	0.4	0.2	0.3	0.55	
b* color	6.9	7.8	9.3	6.1	7.5	8.31	*
p-toluic	326	434	715	160	304	511	*
Metals	7.7	8.4	9.7	3.3	15.5	37.1	
PSD							
60 Mesh	%	19.9		%	22.4		*
80 mesh	%	18.3		%	15.9		*
100 mesh	%	8.2		%	4.5		*
200 mesh	%	32.7		%	29.4		*
325 mesh	%	11.9		%	13.0		*
> 325 mesh	%	9.0		%	14.8		*

* Historical data

As shown, quality data for the solid product coming out from the post oxidizer are similar to CTA data from dryer and the small differences can be due to either sampling procedure or differences in washing procedure, or both. As a matter of fact, slurry in the plant gradually flashed from 21 to 0.5 bara and then washed at high temperature while sampled slurry from the post oxidizer was flashed from 11 to 1 bara and washed according to the procedure described in the experimental section. Moreover, considering that after the post oxidizer, in conventional technology, two more crystallizers are installed, it can be concluded that these pieces of equipment have limited, or even negligible, effect on the final product quality.



APPENDIX 1

In the following the analytical results are reported. For the sake of clarity results relevant to the reactor effluent solid part are reported in Table I, results concerning the reactor effluent liquid part in Table II. Data indicated with A after the date are relevant to R1301A, while data indicated with B after the date are relevant to R1301B. Results concerning the post oxidizer effluent solid part are reported in Table III, results concerning the post oxidizer effluent liquid part in Table IV Analytical method details are reported in [13].



Table I. All data are in expressed in ppm

Reactor Effluent – solid	13/07/04A	09/08/04A	12/08/04A	18/08/04A	02/09/04A	09/09/04A	10/09/04A	04/10/04A	07/10/04A	12/10/04A
1,2,4-Benzene tricarboxylic acid	500		394.78	545.29	607.62	1051	990.67	1018	908.67	739.63
1,2,3-Benzene tricarboxylic acid		694.1								
4-(Hydroxymethyl) benzoic acid	9.75	9.99	14.5	13	15.88	19.24	45.15	22.1	23.9	9.93
Terephthalic acid	15910	15710	15400	15370	16900	1768	16040	15950	15120	13320
4-Carboxybenzaldehyde	6700	5500	5300	5000	4000	3800	8750	6800	7500	4800
Benzoic acid	320	408.57	260.43	195.34	136.71	712.62	570.8	921.26	541.25	266.57
Benzophenone-4,4-Dicarboxylic acid	13.87	13.56		12.1		12.44		41.29		
o-Toluic acid										
p-Toluic acid	1659	1436	792.22	856.15	1745	2854	3506	2203	2558	1055
Diphenylmethane-4,4-Dicarboxylic acid										
Bromo toluic acid										
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	143	119.1	121.82	147.64	139.1	355.38	116.82	149.54	301.48	237.65
Anthracene-9,10-Dicarboxylic acid	16.17	20.68	18.44	36.4	14.89	18.33	18	2.54	29	7.55
% TR 340 nm	17.04	9.84	16.46	4.66	8.69	1.1	3.5	4.8	0	13.21
% TR 400 nm	82.56	77.17	82.14	70.41	76.13	60.64	69	70.7	63.33	77.93
Co	0.53	1.01	29.8	1.2	5.15	11.7	14.05	4.95	8.7	6
Mn	1.81	4.34	67.2	3.8	2.52	5.45	7	1.29	4.2	2.79
Fe	0.54	0.54	2	0.97	1.11	1.38	1.56	0.43	1.7	1.26
L*	98.14	97.61	98.31	98.41	97.85	97.7	97.53	96.7	97.73	98.46
a*	-2.62	-2.84	-2.45	-2.97	-2.92	-3.14	-2.37	-2.88	-2.95	-2.66
b*	8.45	9.42	8.66	10.1	9.82	11.1	9.03	10.51	10.49	9.14

Table I - continue

Reactor Effluent - solid	14/10/04A	18/10/04A	12/11/04A	23/11/04A	25/11/04A	13/07/04B	09/08/04B	12/08/04B	18/08/04B	02/09/04B
1,2,4-Benzene tricarboxylic acid	895.4	793.1	1073	958.64	1274	698	597.32	491.57	392.71	424.73
1,2,3-Benzene tricarboxylic acid		462.65								
4-(Hydroxymethyl) benzoic acid	20.25	17.45	20.47	14.47	24.7	8.8	66.77	8.83		11.63
Terephthalic acid	15920	15720	13680	13520	15100	13750	15640	16130	15390	15730
4-Carboxybenzaldehyde	9500	5300	7250	5000	4500	5700	5700	4700	4250	8200
Benzoic acid	345.3	488.77	564.37	619.44	626.6	484	268.67	335.95	199.06	169.11
Benzophenone-4,4-Dicarboxylic acid						14.15	9.38	8.84	12.48	
o-Toluic acid										
p-Toluic acid	2586	1623	2536	1756	3097	1793	1228	1018	603.5	1533
Diphenylmethane-4,4-Dicarboxylic acid								8.091		
Bromo toluic acid								64.2		
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	393.2	410.5	260.28	304.97	2.85	109.96	119.08	122.5	124.11	139.26
Anthracene-9,10-Dicarboxylic acid	20.53	22.08	34.12	14.4	8.76	21	20.68	40.89	23.72	21.03
% TR 340 nm	1.855	6.05	1.36	1.37	0	16.46	26.66	34.08	34.28	22.75
% TR 400 nm	64.92	70.74	64.8	63.1	40.86	82.44	89.54	90.46	87.77	85.84
Co	11.95	2.5	3.68	4.75	3.8	0.98	2.27	2.94	0.8	5.25
Mn	6.1	1.14	7.3	9.9	9.25	3.26	6.9	5.7	3.2	2.69
Fe	1.1	0.58	1.1	0.47	1.62	0.53	0.62	0.68	0.8	0.74
L*	97.88	97.21	96.58	97.33	96.45	97.85	98.34	98.55	98.25	98.35
a*	-3.2	-3.1	-3.15	-2.63	-3.47	-2.55	-2.04	-1.85	-2.34	-2.19
b*	10.88	10.04	10.7	9.96	12.8	8.67	7.29	6.64	7.98	7.53

Table I - continue

Reactor Effluent - solid	09/09/04B	10/09/04B	04/10/04B	07/10/04B	12/10/04B	14/10/04B	18/10/04B	12/11/04B	23/11/04B	25/11/04B
1,2,4-Benzenetricarboxylic acid	700.6	627.24	947.27	632.3	806.23	665.83	550.3	473.75	945.68	883.44
1,2,3-Benzenetricarboxylic acid										
4-(Hydroxymethyl) benzoic acid	10.57	15.36	29.3	10.82	9.05	10.98	13.74	7.98	15.18	25.72
Terephthalic acid	16880	15780	15650	15870	12900	15750	15190	15350	13640	16800
4-Carboxybenzaldehyde	2900	6750	7800	5500	4800	5300	5300	4750	3750	3750
Benzoic acid	350.8	292	1475	354.37	333.94	312.59	243.74	63.53	542	383.23
Benzophenone-4,4-Dicarboxylic acid		21.39								
o-Toluic acid										
p-Toluic acid	1723	1471	3430	1500	1313	1404	1472	892.39	1584	1418
Diphenylmethane-4,4-Dicarboxylic acid										
Bromo toluic acid										
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	138.5	154	90.2	171.51	179.1	189.9	194.14	238.74	226.1	16.93
Anthracene-9,10-Dicarboxylic acid	16.29	24.17	18.32	34.05	33.65	31.94	31.64	32.85	13.51	16.21
% TR 340 nm	19.1	21.9	12.4	24	25.71	21.53	26.6	32.15	9.66	13.21
% TR 400 nm	81.61	82.9	80.5	85.45	84.82	83.65	85.83	87.8	76.32	80
Co	16.85	7.6	11.2	8.3	8.65	10.65	5.7	2.18	2.59	3.25
Mn	8.15	3.7	6.7	4.2	4.17	5.3	2.8	4.55	5.5	7
Fe	1	2.7	1	2.8	0.72	0.8	0.6	0.6	0.5	0.55
L*	97.57	97.54	97.11	97.64	97.72	97.59	97.11	97.45	97.07	97.16
a*	-2.23	-1.81	-2.13	-2.15	-2.18	-2.53	-2.12	-2.33	-2.65	-3.2
b*	8.17	7.68	8.14	7.88	8.26	9.02	7.08	7.78	9.75	10.49

Table II. All data are expressed in ppm, except for acetic acid and water that are in %

Reactor Effluent - liquid	13/07/04A	09/08/04A	12/08/04A	18/08/04A	02/09/04A	09/09/04A	10/09/04A	04/10/04A	07/10/04A	12/10/04A
% Acetic acid	87.36	83.64	92.9	87.86	88.67	89	90.17	86.65	90.63	90.3
% Water	12.44	13.25	14.26	13.71	12.02	10.5	10.98	12.63	12.57	11.69
Br free	119.3	123.1	99.54	144.1	147.7	122.4	100.55	154.18	129.59	131
Bromine	769.9	889.2	851.72	863.1	735.9	779.1	692.78	726.47	691.35	779.71
Cobalt	223.6	273.1	262.78	277.2	238.3	266.2	245.86	232.75	217.32	237.24
Manganese	344.9	422.7	533.82	547.1	474.3	533.2	458.18	457.33	437.61	469.22
p-xylene	45570							699.06	1343	986.1
1,2,4-Benzenetricarboxylic acid	970.6	1073	767.98	1210	845.1	1019	658.37	826.67	967.86	1112
1,2,3-Benzenetricarboxylic acid		31.68								
4-(Hydroxymethyl) benzoic acid		222.4	192.72	192.3	200.9	206	193.51	131.88	256.23	178.54
Terephthalic acid	749.3	748.8	640.75	731.9	938.9	716.6	579.34	597.85	488.31	682.6
4-Carboxybenzaldehyde	3355	2532	2211	2438	2292	2231	2380	1570	2570	2456
Benzoic acid	14320	15120	13440	9356	9064	11350	9574	12760	11780	11060
Benzophenone-4,4-Dicarboxylic acid	646	10.74	10.03	11.74	11.94	6.22	6.22	11.67	18.82	6.831
o-Toluic acid										
p-Toluic acid	20440	19340	16950	18020	16970	18080	18620	15510	19350	18360
Diphenylmethane-4,4-Dicarboxylic acid	2200	328	2416	2719	1898	2311	1913	2154	2175	2597
Bromo toluic acid	4553	2877	2140	2502	2702	2601	4403	1977	3350	2796
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	15.11	22.08	12.74	49.96	14.98	21.79	10.15	14.65	20.11	12.45
Anthracene-9,10-Dicarboxylic acid	16.46	2.25	1.74	1.8	2.1	7.03	5.41	0.796	3.52	1.37

Table II - continue

Reactor Effluent - liquid	14/10/04A	18/10/04A	12/11/04A	23/11/04A	25/11/04A	13/07/04B	09/08/04B	12/08/04B	18/08/04B	02/09/04B
% Acetic acid	89.15	89.18	87.14	89.59	88.65	89.27	87.38	91.62	87.73	91.09
% Water	14.21	15.75	11.67	13.37	14.73	11.79	11.17	12.48	14.13	10.88
Br free	152.8	154.2	130.91	164.4	130.2	76.71	120.36	88.81	150.86	143.85
Bromine	772.1	737.7	658.32	772.1	605	723.7	932.14	870.09	927.66	778.79
Cobalt	223.7	211.9	205.57	212.4	176	219.2	282.69	260.33	308.43	247.96
Manganese	450.2	431.7	407.24	422.9	333.2	334.5	445.81	534.09	603.74	489.52
p-xylene	2733	518.6	1122	704.5	1244	20500				
1,2,4-Benzenetricarboxylic acid	1236	881	1043	1038	1626	532.3	585.94	371.55	742.64	579.89
1,2,3-Benzenetricarboxylic acid	15.36				3.54			17.78		
4-(Hydroxymethyl) benzoic acid	454.8	127.1	229.21	233.4	156.2	148.8	207.15	132.12	117.98	293.26
Terephthalic acid	574	533.6	619.7	491	531.5	559.5	597.46	635.49	753.88	924.33
4-Carboxybenzaldehyde	3491	1302	2509	1418	1696	1948	2707	2155	1995	3305
Benzoic acid	12480	8.8	10850	9358	10550	12340	11080	13080	11780	8705
Benzophenone-4,4-Dicarboxylic acid	25.1	9.26	17.31	14.98	24.23	13.58	12.2	7.07	13.1	10.43
p-Toluic acid	2399	13580	20020	14190	17040	17830	19620	17440	16810	19720
Diphenylmethane-4,4-Dicarboxylic acid	2587	2237	2102	2244	2379	1883	2434	2140	2993	1782
Bromo toluic acid	6426	1304	2739	1589	1939	2699	4364	3632	3465	5801
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	8.23	70.56	43	4.61	6.17	13.63	18.13	14.87	19.57	15.93
Anthracene-9,10-Dicarboxylic acid	15.43	1.05	4.02	15.76	1.74	3.32	2.07	1.75	1.8	1.45

Table II - continue

Reactor Effluent - liquid	09/09/04B	10/09/04B	04/10/04B	07/10/04B	12/10/04B	14/10/04B	18/10/04B	12/11/04B	23/11/04B	25/11/04B
% Acetic acid	89.4	89.02	87.05	89.54	89.73	88.04	87.38	88.6	88.6	86.58
% Water	10.5	10.18	13.95	13.48	12.07	14.16	16.95	11.32	12.29	13.24
Br free	104.2	111.6	128.86	127	132	150.6	116.36	136.94	148.36	139.63
Bromine	802.2	790.8	690.45	711.8	782.3	790	644.83	717.28	807.9	746.24
Cobalt	271.3	281.8	212.57	223.6	245.3	232.4	181.9	219.14	225	213.64
Manganese	540.4	563.6	418.79	449.4	493.3	464.8	360.82	436.11	450.23	411.21
p-xylene			2626	2316	1720	2007	3033	2332	1435	2016
1,2,4-Benzenetricarboxylic acid	578.8	578.2	586.44	805.5		844.9		653	718.35	1027
1,2,3-Benzenetricarboxylic acid							26.82			
4-(Hydroxymethyl) benzoic acid	142.8	168.4	187.2	190	117.4	162.1	123.73	148.41	70.95	139.52
Terephthalic acid	675.1	631	631.37	569.8	650.5	550	443.52	543.48	549.76	588.08
4-Carboxybenzaldehyde	2376	2600	2465	2774	2078	2340	1715	2348	1281	2058
Benzoic acid	10060	10880	11210	13320	15560	12180	7401	10370	10250	11350
Benzophenone-4,4-Dicarboxylic acid	8.22	9.42	10.42	19.6	5.63	17.91	6.45	11.74	14.14	21.72
p-Toluic acid	18470	18720	18610	19220	17820	18.73	16770	18790	14110	19330
Diphenylmethane-4,4-Dicarboxylic acid	2072	2199	1869	2155	2127	2302	1733	1863	2419	2536
Bromo toluic acid	3442	3595	4297	4841	2490	3924	3388	3275	2446	2994
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	6.6	13.1	6.3	25.26	22.6	20.79	15.558	3.97	4.17	10.53
Anthracene-9,10-Dicarboxylic acid	0.99	3.9	1.48	2.6	2.2	1.93	0.8	6.06	5.04	1.74

Table III. All data are in expressed in ppm

Post oxidizer effluent- solid	12/7/2004	13/07/04	9/8/2004	18/08/04	2/9/2004	9/9/2004	10/9/2004	4/10/2004
p-xylene		46520						
1,2,4-Benzenetricarboxylic acid	891.83	256.72	258	318.3	205.0	281.9	270.9	275.4
1,2,3-Benzenetricarboxylic acid								
4-(Hydroxymethyl) benzoic acid	17.63							9.0
Terephthalic acid	15560	13920	15800	15600.0	16220.0	16190.0	15820.0	15830.0
4-Carboxybenzaldehyde	0.35	0.35	0.3	0.3	0.4	0.4	0.4	0.3
Benzoic acid	760.77	206	159.37	187.8	131.5	181.2	170.1	464.4
Benzophenone-4,4-Dicarboxylic acid	17.62	11.86	8.82	11.4				
o-Toluic acid								
p-Toluic acid	2184	519.4	354.7	359.9	425.9	437.5	453.6	714.6
Diphenylmethane-4,4-Dicarboxylic acid								
Bromo toluic acid								
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid		282	118.8	121.8	138.4	133.7	151.5	183.4
Anthracene-9,10-Dicarboxylic acid	105.77		21.81	22.5	21.1	18.0	24.2	31.7
% TR 340 nm	21.17	36.16	37.16	17.8	22.0	30.4	35.6	39.2
% TR 400 nm	88.51	89.1	90.94	80.7	85.9	83.7	87.7	89.7
Co	1.73	2.64	1.62	2.4	6.5	12.8	13.3	7.3
Mn	4.73	6.05	6.05	9.3	3.0	5.2	6.8	3.4
Fe	1.47	5.5	5.5	1.1	0.8	0.9	2.4	0.5
L*	98.14	98.49	98.6	98.2	98.7	97.8	98.3	98.4
a*	-2.17	-2.11	-2.19	-2.5	-2.1	-2.1	-1.9	-2.1
b*	7.4	7.09	7.05	9.0	6.9	7.6	7.1	7.3

Table III - continue

Post oxidizer - solid	7/10/2004	12/10/2004	14/10/2004	18/10/2004	12/11/2004	23/11/2004	25/11/2004
p-xylene							
1,2,4-Benzenetricarboxylic acid	300.7	360.2	393.9	237.7	303.1	315.9	455.1
1,2,3-Benzenetricarboxylic acid							
4-(Hydroxymethyl) benzoic acid	6.3		8.3	5.5	6.3		14.6
Terephthalic acid	15880.0	13170.0	15260.0	15910.0	15470.0	13340.0	15170.0
4-Carboxybenzaldehyde	0.3	0.3	0.4	0.3	0.3	0.2	0.2
Benzoic acid	185.4	152.0	155.9	141.1	146.6	121.8	122.1
Benzophenone-4,4-Dicarboxylic acid							
O-Toluic acid							
P-Toluic acid	418.1	358.0	522.4		362.1	326.2	472.0
Diphenylmethane-4,4-Dicarboxylic acid							
Bromo toluic acid							
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	171.6	166.4	191.1	234.2	241.6	226.9	13.2
Anthracene-9,10-Dicarboxylic acid	19.3	16.5	33.2	31.7	18.2	29.5	17.2
% TR 340 nm	31.4	34.7	21.5	37.0	26.6	24.2	8.4
% TR 400 nm	86.8	85.9	81.9	87.0	85.2	84.0	76.4
Co	7.9	11.0	24.6	6.1	4.8	3.1	4.8
Mn	3.9	4.0	10.9	2.9	8.9	6.4	20.0
Fe	1.4	0.6	1.6	0.4	1.2	0.5	1.3
L*	98.2	97.8	98.2	98.3	97.6	97.7	97.2
a*	-2.0	-2.0	-2.4	-2.1	-2.3	-2.4	-2.5
b*	7.5	7.4	8.4	7.5	7.7	8.8	9.3

Table IV. All data are expressed in ppm, except for acetic acid and water that are in %

Post oxidizer - liquid	12/7/2004	13/07/04	9/8/2004	18/08/04	2/9/2004	9/9/2004	10/9/2004	4/10/2004
% Acetic acid	90.4	91.41	95.39	91.29	92.08	91	91.7	88.7
% Water	10.4	11.5	10.8	10.2	10.4	8.5	5.3	11.3
Br free	488	499.7	518.3	605.7	543.3	440.2	435.1	592.7
Bromine	1059	1043	1139	1271	1119	1158	1118	1116
Cobalt	318.9	320	381.6	437.3	368.5	410.5	409.6	341.6
Manganese	513.2	531.8	625.8	849	743.9	829.2	823.9	690.3
p-xylene	18760	14490						
1,2,4-Benzenetricarboxylic acid	886.9	1089		1025		1145	960.1	1172
1,2,3-Benzenetricarboxylic acid	57.1			43.14				
4-(Hydroxymethyl) benzoic acid	251.9	339.5	316.6	397.2	182.1	293.5	301.3	208.7
Terephthalic acid	402.4	740.9	746.9	652.5	620.2	748.4	562.6	596
4-Carboxybenzaldehyde	293.8	420.9	380.3	307.4	404.2	358	378.6	380.7
Benzoic acid	14020	17710	16250	12970	12360	15530	14220	16750
Benzophenone-4,4-Dicarboxylic acid	24.91	33.35	3792	29.16	39.22	35.84	28.27	29.89
o-Toluic acid							66.14	
p-Toluic acid	2605	4242	4224	3730	4121	3975	2192	3721
Diphenylmethane-4,4-Dicarboxylic acid	1046	336.9	1596	809.3	238.3	319.4	290.5	1284
Bromo toluic acid	30.88	1426	40.22	48.81	1521	2039	2039	34.54
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	28.15	37.16		27.45	5.81	9.93	25.95	1.788
Anthracene-9,10-Dicarboxylic acid	2.65	3.8	10	1.8	11.27	5	4.28	1.778

Table IV - continue

Post oxidizer - liquid	7/10/2004	12/10/2004	14/10/2004	18/10/2004	12/11/2004	23/11/2004	25/11/2004
% Acetic acid	91.31	93.24	90.96	90.14	90.52	90.53	92.67
% Water	11.0	10.3	12.9	13.6	9.6	10.6	10.7
Br free	533	514.18	618.28	631.31	561.47	622.54	594.18
Bromine	1067.77	1124.84	1265	1142.34	1049.24	1168	1121
Cobalt	336.64	349.27	367.4	317.64	322.21	319.9	326.69
Manganese	685.6	699	744.55	644.72	648.77	644.15	623.57
p-xylene	448.16		439.25		331.22	264.32	484.12
1,2,4-Benzenetricarboxylic acid	1616		1656				
1,2,3-Benzenetricarboxylic acid			236.18				
4-(Hydroxymethyl) benzoic acid	450.64	226.52	240.49	204.54	207.43	214.68	505.33
Terephthalic acid	691.21	541.66	466.58	494.29	619.3	538.38	753.78
4-Carboxybenzaldehyde	465.82	329.52	548.88	317.27	400.78	217.42	415.46
Benzoic acid	21390	1428	17260	12570	15910	13050	17940
Benzophenone-4,4-Dicarboxylic acid	44.27	27.27	39	28.96	29.29	17.68	24.52
o-Toluic acid							
p-Toluic acid	4448	3535	46.58	2930	3917	17.92	3483
Diphenylmethane-4,4-Dicarboxylic acid	1910	1418	1741	1032	1558	176.7	1472
Bromo toluic acid	61.12	22.44	53.21	19.19	28.41	917.66	6.1
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid	9.43	36.47	42.29	28.67	1.84	4.83	28.81
Anthracene-9,10-Dicarboxylic acid	4.7	2.3	2.19	1.512	16.28	8.28	2.45

APPENDIX 2

In the following the material balance for each run is reported. Stream number and plant scheme are reported in figure 1. In the top left corner of each table the date of the relevant test is reported. In the body of the table “-“ means that the corresponding compound is not present, N/A that the analysis is not available, but the compound can be present. All figures are in kg/h.

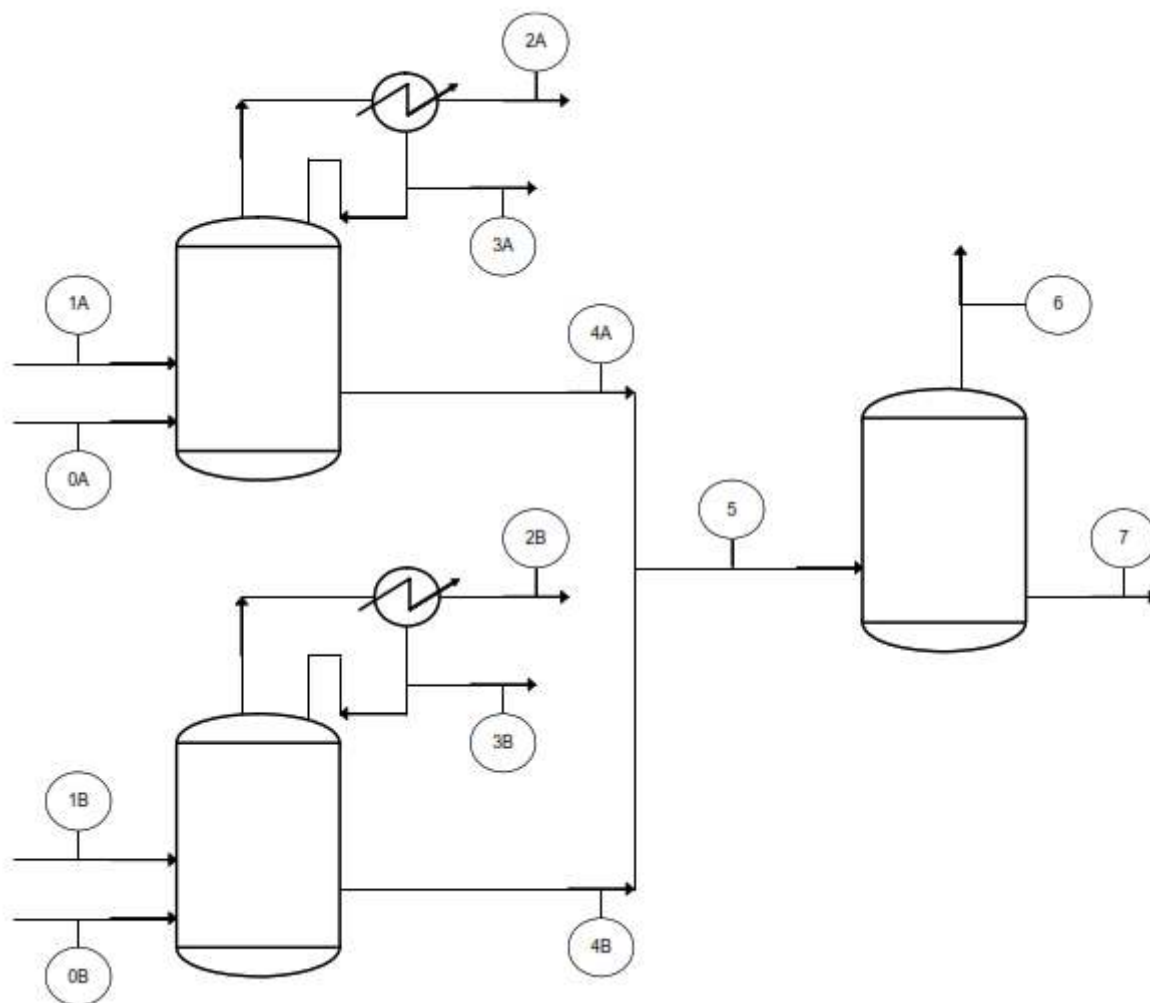


Figure 17. Plant scheme.



13/07/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6839	6861	0	0	N/A	N/A	-	932	-	802	-	1734	-	-	754
acetic acid			22896	22969	0	0	4641	4340	-	17866	-	34907	-	52773	5204	-	47569
H ₂ O			1833	1839	48	48	1559	1860	-	2544	-	4610	-	7154	-	-	5979
Br			18	18	-	-	-	-	-	16	-	28	-	44	-	-	54
Br ⁻			11	11	-	-	-	-	-	2	-	3	-	5	-	-	26
Co			6	6	-	-	-	-	N/A	5	N/A	9	N/A	13	-	N/A	17
Mn			8	8	-	-	-	-	N/A	7	N/A	13	N/A	20	-	N/A	28
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	5	20	7	21	13	41	-	5	57
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	0	0	6	0	6	-	0	18
Terephthalic acid			N/A	N/A	-	-	-	-	10435	15	10473	22	20908	37	-	20999	39
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	71	69	60	76	131	145	-	74	22
Benzoic acid			N/A	N/A	-	-	-	-	3	293	5	483	8	775	-	4	922
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	13	0	1	0	14	-	0	2
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	17	418	19	697	36	1115	-	11	221
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	45	0	74	0	119	-	0	18
Bromo toluic acid			N/A	N/A	-	-	-	-	0	93	0	106	0	199	-	0	74
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	2	0	1	1	3	1	-	6	2
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7658	7710	-	-	1268	1268	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	503	565	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	74	104	-	-	-	-	-	-	-	-	-	-	-
N ₂	24572	24740	-	-	24572	24572	-	-	-	-	-	-	-	-	-	-	-
TOTAL	32230	32450	31600	31700	26465	26556	6200	6200	10533	22336	10567	41854	21100	64190	5204	21100	55774

* estimated with Aspen plus^o

09/08/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6961	6939	0	0	N/A	N/A	-	0	-	0	-	0	-	-	0
acetic acid			23499	23426	0	0	4707	5235	-	18361	-	31674	-	50035	4220	-	45815
H ₂ O			1605	1600	50	50	2093	2165	-	2909	-	4049	-	6958	-	-	5163
Br			19	19	-	-	-	-	-	20	-	34	-	53	-	-	55
Br ⁻			14	14	-	-	-	-	-	3	-	4	-	7	-	-	25
Co			6	6	-	-	-	-	N/A	6	N/A	10	N/A	16	-	N/A	18
Mn			10	10	-	-	-	-	N/A	9	N/A	16	N/A	25	-	N/A	30
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	24	6	21	6	45	-	6	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	1	0	0	0	1	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	5	1	8	1	12	-	0	15
Terephthalic acid			N/A	N/A	-	-	-	-	10686	16	10647	22	21333	38	-	21416	36
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	59	56	61	98	120	154	-	65	18
Benzoic acid			N/A	N/A	-	-	-	-	4	332	3	402	7	734	-	3	780
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	1	-	0	182
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	15	425	13	711	29	1136	-	8	203
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	7	0	88	0	95	-	0	77
Bromo toluic acid			N/A	N/A	-	-	-	-	0	63	0	158	0	221	-	0	2
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	1	0	1	1	3	1	-	3	0
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7983	7603	-	-	1248	1233	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	565	468	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	77	87	-	-	-	-	-	-	-	-	-	-	-
N ₂	25616	24396	-	-	25617	25617	-	-	-	-	-	-	-	-	-	-	-
TOTAL	33600	32000	32100	32000	27557	27455	6800	7400	10767	22233	10733	37292	21500	59525	4220	21500	52395

* estimated with Aspen plus^o

12/08/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6616	6484	0	0	N/A	N/A	-	0	-	0	-	0	-	-	
acetic acid			21781	21345	0	0	6900	6900	-	14553	-	28360	-	42912	3440	-	
H ₂ O			1575	1544	47	47	0	0	-	2234	-	3863	-	6097	-	-	
Br			14	14	-	-	-	-	-	13	-	27	-	40	-	-	
Br ⁻			11	11	-	-	-	-	-	2	-	3	-	4	-	-	
Co			4	4	-	-	-	-	N/A	4	N/A	8	N/A	12	-	N/A	
Mn			9	9	-	-	-	-	N/A	8	N/A	17	N/A	25	-	N/A	
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	4	12	5	12	9	24	-	0	
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	1	0	1	-	0	
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	3	0	4	0	7	-	0	
Terephthalic acid			N/A	N/A	-	-	-	-	10132	10	9930	20	20062	30	-	20200	
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	54	35	47	67	101	101	-	0	
Benzoic acid			N/A	N/A	-	-	-	-	3	211	3	405	6	615	-	0	
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	
p-Toluic acid			N/A	N/A	-	-	-	-	8	266	10	540	18	805	-	0	
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	38	0	66	0	104	-	0	
Bromo toluic acid			N/A	N/A	-	-	-	-	0	34	1	112	1	146	-	0	
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	1	0	1	0	2	1	-	0	
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	1	0	-	0	
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	
O ₂	7524	7175	-	-	1245	1190	-	-	-	-	-	-	-	-	-	-	
CO ₂			-	-	434	483	-	-	-	-	-	-	-	-	-	-	
CO			-	-	73	87	-	-	-	-	-	-	-	-	-	-	
N ₂	24145	23024	-	-	24145	24145	-	-	-	-	-	-	-	-	-	-	
TOTAL	31670	30200	30000	29400	25944	25953	6900	6900	10202	17420	9998	33501	20200	50920	3440	20200	

* estimated with Aspen plus^o

18/08/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			7250	6450	0	0	N/A	N/A	-	0	-	0	-	0	-	-	0
acetic acid			23008	20469	0	0	4828	5000	-	17810	-	31410	-	49220	3800	-	45420
H ₂ O			1968	1751	47	47	2372	2200	-	2779	-	5059	-	7838	-	-	5065
Br			17	15	-	-	-	-	-	17	-	33	-	51	-	-	63
Br ⁻			15	13	-	-	-	-	-	3	-	5	-	8	-	-	30
Co			5	5	-	-	-	-	N/A	6	N/A	11	N/A	17	-	N/A	22
Mn			12	11	-	-	-	-	N/A	11	N/A	22	N/A	33	-	N/A	42
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	6	25	4	27	10	51	-	7	51
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	2
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	4	0	4	0	8	-	0	20
Terephthalic acid			N/A	N/A	-	-	-	-	11090	15	9878	27	20968	42	-	21020	32
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	56	49	42	71	98	121	-	58	15
Benzoic acid			N/A	N/A	-	-	-	-	2	190	2	422	4	611	-	4	645
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	1	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	10	365	6	602	16	967	-	8	186
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	55	0	107	0	162	-	0	40
Bromo toluic acid			N/A	N/A	-	-	-	-	0	51	0	124	0	175	-	0	2
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	2	1	1	1	3	2	-	3	1
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	1	0	-	0	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7603	7484	-	-	1258	1258	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	473	554	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	73	88	-	-	-	-	-	-	-	-	-	-	-
N ₂	24396	24015	-	-	24397	24397	-	-	-	-	-	-	-	-	-	-	-
TOTAL	32000	31500	32260	28700	26249	26344	7200	7200	11166	21378	9934	37920	21100	59298	3800	21100	51609

* estimated with Aspen plus^o

02/09/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6556	6644	0	0	N/A	N/A	-	0	-	0	-	0	-	-	0
acetic acid			20799	21080	0	0	5062	5109	-	15363	-	30587	-	45950	3000	-	42950
H ₂ O			2220	2250	48	48	1938	1891	-	2083	-	3653	-	5736	-	-	4828
Br			13	13	-	-	-	-	-	13	-	26	-	39	-	-	52
Br ⁻			9	9	-	-	-	-	-	3	-	5	-	7	-	-	25
Co			4	4	-	-	-	-	N/A	4	N/A	8	N/A	12	-	N/A	17
Mn			9	9	-	-	-	-	N/A	8	N/A	16	N/A	25	-	N/A	35
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	6	15	4	19	11	34	-	4	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	3	0	10	0	13	-	0	8
Terephthalic acid			N/A	N/A	-	-	-	-	10113	16	10210	31	20324	47	-	20404	29
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	41	40	85	111	125	151	-	77	19
Benzoic acid			N/A	N/A	-	-	-	-	1	157	2	292	3	449	-	3	577
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	1	-	0	2
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	18	294	16	662	34	956	-	9	192
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	33	0	60	0	93	-	0	11
Bromo toluic acid			N/A	N/A	-	-	-	-	0	47	0	195	0	242	-	0	71
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	1	0	1	1	3	1	-	3	0
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	1
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7722	7555	-	-	1278	1278	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	586	508	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	75	87	-	-	-	-	-	-	-	-	-	-	-
N ₂	24778	24244	-	-	24778	24778	-	-	-	-	-	-	-	-	-	-	-
TOTAL	32500	31800	29600	30000	26764	26699	7000	7000	10181	18076	10319	35673	20500	53748	3000	20500	48791

* estimated with Aspen plus^o

09/09/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			7692	6708	0	0	N/A	N/A	-	0	-	0	-	0	-	-	0
acetic acid			25368	22123	0	0	5183	4958	-	19829	-	32110	-	51939	4200	-	47739
H ₂ O			1307	1140	53	53	1817	1542	-	2339	-	3771	-	6111	-	-	4480
Br			16	14	-	-	-	-	-	17	-	29	-	46	-	-	61
Br ⁻			13	11	-	-	-	-	-	3	-	4	-	6	-	-	23
Co			5	5	-	-	-	-	N/A	6	N/A	10	N/A	16	-	N/A	22
Mn			12	10	-	-	-	-	N/A	12	N/A	19	N/A	31	-	N/A	44
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	12	23	7	21	20	43	-	6	60
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	5	0	5	0	10	-	0	15
Terephthalic acid			N/A	N/A	-	-	-	-	11754	16	10281	24	22035	40	-	22099	39
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	45	50	30	85	75	135	-	78	19
Benzoic acid			N/A	N/A	-	-	-	-	8	253	4	361	12	614	-	4	815
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	2
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	34	403	18	663	52	1066	-	10	209
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	51	0	74	0	126	-	0	17
Bromo toluic acid			N/A	N/A	-	-	-	-	0	58	0	124	0	182	-	0	107
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	4	0	1	0	6	1	-	3	1
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	8553	7389	-	-	1321	1416	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	476	519	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	83	28	-	-	-	-	-	-	-	-	-	-	-
N ₂	27446	23710	-	-	27446	27446	-	-	-	-	-	-	-	-	-	-	-
TOTAL	36000	31100	34400	30000	29379	29462	7000	6500	11858	23063	10342	37298	22200	60361	4200	22200	53628

* estimated with Aspen plus^o

10/09/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			7706	6694	0	0	N/A	N/A	-	0	-	0	-	0	-	-	0
acetic acid			25824	22430	0	0	5381	4969	-	20110	-	32240	-	52350	4100	-	48250
H ₂ O			1435	1246	53	53	1419	1531	-	2449	-	3687	-	6136	-	-	2768
Br			17	15	-	-	-	-	-	15	-	29	-	44	-	-	59
Br ⁻			13	11	-	-	-	-	-	2	-	4	-	6	-	-	23
Co			5	5	-	-	-	-	N/A	5	N/A	10	N/A	16	-	N/A	22
Mn			12	11	-	-	-	-	N/A	10	N/A	20	N/A	31	-	N/A	43
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	12	15	6	21	18	36	-	6	51
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	1	4	0	6	1	10	-	0	16
Terephthalic acid			N/A	N/A	-	-	-	-	11714	13	10223	23	21937	36	-	22082	30
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	104	53	70	94	174	147	-	94	20
Benzoic acid			N/A	N/A	-	-	-	-	7	214	3	394	10	608	-	4	748
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	3
p-Toluic acid			N/A	N/A	-	-	-	-	42	415	15	678	57	1093	-	10	115
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	43	0	80	0	122	-	0	15
Bromo toluic acid			N/A	N/A	-	-	-	-	0	98	0	130	0	228	-	0	107
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	1	0	2	0	3	1	-	3	1
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	1	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	8458	7484	-	-	1307	1307	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	458	513	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	82	82	-	-	-	-	-	-	-	-	-	-	-
N ₂	27141	24015	-	-	27141	27141	-	-	-	-	-	-	-	-	-	-	-
TOTAL	35600	31500	35000	30400	29040	29096	6800	6500	11881	23445	10319	37413	22200	60858	4100	22200	52250

* estimated with Aspen plus^o

04/10/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6738	6962	0	0	N/A	N/A	-	14	-	92	-	106	-	-	0
acetic acid			22151	22890	0	0	4933	4891	-	16871	-	30479	-	47350	3022	-	44328
H ₂ O			1978	2044	48	48	1867	1909	-	2459	-	4884	-	7343	-	-	5632
Br			18	19	-	-	-	-	-	14	-	24	-	38	-	-	56
Br ⁻			13	13	-	-	-	-	-	3	-	5	-	8	-	-	30
Co			5	5	-	-	-	-	N/A	5	N/A	7	N/A	12	-	N/A	17
Mn			10	11	-	-	-	-	N/A	9	N/A	15	N/A	24	-	N/A	34
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	11	16	10	21	21	37	-	6	59
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	3	0	7	1	9	-	0	10
Terephthalic acid			N/A	N/A	-	-	-	-	10456	12	10776	22	21231	34	-	21399	30
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	72	31	85	86	157	117	-	65	19
Benzoic acid			N/A	N/A	-	-	-	-	10	248	16	392	26	641	-	10	837
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	1	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	23	302	37	652	61	954	-	15	186
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	42	0	65	0	107	-	0	64
Bromo toluic acid			N/A	N/A	-	-	-	-	0	38	0	150	0	189	-	0	2
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	2	0	1	0	3	1	-	4	0
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	1	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7722	7555	-	-	1193	1193	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	461	434	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	75	N/A	-	-	-	-	-	-	-	-	-	-	-
N ₂	24778	24244	-	-	24778	24778	-	-	-	-	-	-	-	-	-	-	-
TOTAL	32500	31800	30900	31930	26554	26452	6800	6800	10574	20064	10926	36898	21500	56961	3022	21500	51276

* estimated with Aspen plus^o

07/10/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6659	5841	0	0	N/A	N/A	-	24	-	93	-	118	-	-	24
acetic acid			21524	18878	0	0	4777	3809	-	16404	-	36081	-	52485	3200	-	49285
H ₂ O			1881	1650	47	47	2073	1591	-	2275	-	5432	-	7707	-	-	5927
Br			20	17	-	-	-	-	-	13	-	29	-	41	-	-	58
Br ⁻			14	12	-	-	-	-	-	2	-	5	-	7	-	-	29
Co			5	5	-	-	-	-	N/A	4	N/A	9	N/A	13	-	N/A	18
Mn			11	10	-	-	-	-	N/A	8	N/A	18	N/A	26	-	N/A	37
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	9	18	6	32	15	50	-	6	87
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	5	0	8	0	12	-	0	24
Terephthalic acid			N/A	N/A	-	-	-	-	10265	9	9037	23	19302	32	-	19420	37
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	78	47	50	112	128	158	-	59	25
Benzoic acid			N/A	N/A	-	-	-	-	6	213	3	537	9	750	-	4	1155
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	1	0	1	-	0	2
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	27	350	14	774	40	1125	-	8	240
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	39	0	87	0	126	-	0	103
Bromo toluic acid			N/A	N/A	-	-	-	-	0	61	0	195	0	256	-	0	3
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	3	0	2	1	5	1	-	3	1
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	1	0	-	0	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7555	6652	-	-	1167	1167	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	504	730	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	73	97	-	-	-	-	-	-	-	-	-	-	-
N ₂	24244	21347	-	-	24244	24244	-	-	-	-	-	-	-	-	-	-	-
TOTAL	31800	28000	30100	26400	26036	26286	6850	5400	10388	19470	9112	43432	19500	62902	3200	19500	57027

* estimated with Aspen plus^o

12/10/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6685	6695	0	0	N/A	N/A	-	19	-	59	-	78	-	-	0
acetic acid			22393	22424	0	0	4698	4851	-	17405	-	30763	-	48168	3400	-	44768
H ₂ O			1484	1486	47	47	1951	1919	-	2253	-	4138	-	6391	-	-	4965
Br			19	19	-	-	-	-	-	15	-	27	-	42	-	-	54
Br ⁻			13	13	-	-	-	-	-	3	-	5	-	7	-	-	25
Co			5	5	-	-	-	-	N/A	5	N/A	8	N/A	13	-	N/A	17
Mn			11	11	-	-	-	-	N/A	9	N/A	17	N/A	26	-	N/A	34
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	8	21	8	0	16	21	-	7	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	3	0	4	0	7	-	0	11
Terephthalic acid			N/A	N/A	-	-	-	-	10170	13	10181	22	20350	35	-	20421	26
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	49	47	49	71	98	119	-	57	16
Benzoic acid			N/A	N/A	-	-	-	-	3	213	3	533	6	747	-	3	69
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	11	354	13	611	24	965	-	7	170
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	50	0	73	0	123	-	0	68
Bromo toluic acid			N/A	N/A	-	-	-	-	0	54	0	85	0	139	-	0	1
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	2	0	2	1	4	1	-	3	2
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7549	7656	-	-	1127	2574	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	443	458	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	73	73	-	-	-	-	-	-	-	-	-	-	-
N ₂	24223	24569	-	-	24223	24223	-	-	-	-	-	-	-	-	-	-	-
TOTAL	31772	32226	30597	30640	25913	27375	6649	6770	10243	20462	10257	36413	20500	56876	3400	20500	50200

* estimated with Aspen plus^o

14/10/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6794	6706	0	0	N/A	N/A	-	52	-	72	-	124	-	-	22
acetic acid			21736	21452	0	0	4672	4778	-	16820	-	31698	-	48518	2700	-	45818
H ₂ O			2035	2008	47	47	2128	2022	-	2681	-	5098	-	7779	-	-	6498
Br			19	19	-	-	-	-	-	15	-	28	-	43	-	-	64
Br ⁻			13	13	-	-	-	-	-	3	-	5	-	8	-	-	31
Co			5	5	-	-	-	-	N/A	4	N/A	8	N/A	13	-	N/A	19
Mn			10	10	-	-	-	-	N/A	8	N/A	17	N/A	25	-	N/A	38
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	9	23	7	30	16	54	-	8	83
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	12
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	9	0	6	0	14	-	0	12
Terephthalic acid			N/A	N/A	-	-	-	-	10424	11	10348	20	20772	31	-	20889	24
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	100	66	55	84	156	150	-	84	28
Benzoic acid			N/A	N/A	-	-	-	-	4	235	3	439	7	674	-	3	869
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	1	0	1	-	0	2
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	27	45	15	1	42	46	-	11	2
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	49	0	83	0	132	-	0	88
Bromo toluic acid			N/A	N/A	-	-	-	-	0	121	0	141	0	263	-	0	3
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	4	0	2	1	6	1	-	4	2
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	1	0	-	1	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7531	7579	-	-	1164	1164	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	400	522	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	73	82	-	-	-	-	-	-	-	-	-	-	-
N ₂	24168	24320	-	-	24168	24168	-	-	-	-	-	-	-	-	-	-	-
TOTAL	31700	31900	30600	30200	25851	25983	6800	6800	10569	20140	10431	37727	21000	57867	2700	21000	53582

* estimated with Aspen plus^o

18/10/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6397	6603	0	0	N/A	N/A	-	9	-	112	-	122	-	-	0
acetic acid			19986	20631	0	0	3727	4124	-	15856	-	32368	-	48224	3100	-	45124
H ₂ O			1776	1833	45	45	1984	2096	-	2800	-	6279	-	9079	-	-	6788
Br			18	18	-	-	-	-	-	13	-	24	-	37	-	-	57
Br ⁻			13	13	-	-	-	-	-	3	-	4	-	7	-	-	32
Co			4	4	-	-	-	-	N/A	4	N/A	7	N/A	11	-	N/A	16
Mn			9	9	-	-	-	-	N/A	8	N/A	13	N/A	21	-	N/A	32
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	8	16	6	0	13	16	-	5	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	1	0	1	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	2	0	5	0	7	-	0	10
Terephthalic acid			N/A	N/A	-	-	-	-	9756	9	10080	16	19836	26	-	19932	25
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	52	23	54	64	106	87	-	55	16
Benzoic acid			N/A	N/A	-	-	-	-	5	0	2	274	7	274	-	3	629
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	16	241	15	621	31	863	-	0	147
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	40	0	64	0	104	-	0	52
Bromo toluic acid			N/A	N/A	-	-	-	-	0	23	0	125	0	149	-	0	1
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	4	1	2	1	6	2	-	5	1
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	1	0	-	1	0
Methane			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	N/A	N/A	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	N/A	N/A	-	-	-	-	-	-	-	-	-	-	-
O ₂	7310	7389	-	-	1129	1129	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	525	551	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	71	N/A	-	-	-	-	-	-	-	-	-	-	-
N ₂	23459	23710	-	-	23459	23459	-	-	-	-	-	-	-	-	-	-	-
TOTAL	30770	31100	28190	29100	25229	25185	5711	6220	9841	19047	10159	39974	20000	59021	3100	20000	52900

* estimated with Aspen plus^o

12/11/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			6545	6655	21	11	N/A	N/A	-	21	-	88	-	109	-	-	17
acetic acid			21518	21877	26	51	4825	5461	-	16285	-	33359	-	49644	3700	-	45944
H ₂ O			1905	1937	46	46	1975	1939	-	2181	-	4262	-	6443	-	-	4878
Br			17	18	-	-	-	-	-	12	-	27	-	39	-	-	53
Br ⁻			13	13	-	-	-	-	-	2	-	5	-	8	-	-	28
Co			5	5	-	-	-	-	N/A	4	N/A	8	N/A	12	-	N/A	16
Mn			9	10	-	-	-	-	N/A	8	N/A	16	N/A	24	-	N/A	33
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	11	19	5	25	15	44	-	6	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	4	0	6	0	10	-	0	11
Terephthalic acid			N/A	N/A	-	-	-	-	9801	12	10018	20	19818	32	-	19928	31
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	72	47	48	88	120	135	-	50	20
Benzoic acid			N/A	N/A	-	-	-	-	6	203	1	390	6	593	-	3	808
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	1	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	25	374	9	707	34	1082	-	7	199
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	39	0	70	0	109	-	0	79
Bromo toluic acid			N/A	N/A	-	-	-	-	0	51	0	123	0	174	-	0	1
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	3	1	2	0	5	1	-	5	0
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	1	0	-	0	1
Methane			-	-	137	1	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	15	1	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	320	21	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	0	0	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	1	0	-	-	-	-	-	-	-	-	-	-	-
O ₂	7460	7698	-	-	1154	1154	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	453	604	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	72	N/A	-	-	-	-	-	-	-	-	-	-	-
N ₂	23939	24701	-	-	23939	23932	-	-	-	-	-	-	-	-	-	-	-
TOTAL	31400	32400	30000	30500	26185	25822	6800	7400	9917	19262	10083	39192	20000	58454	3700	20000	52093

* estimated with Aspen plus ^o

23/11/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			7038	6962	6	3	N/A	N/A	-	13	-	56	-	69	-	-	14
acetic acid			22521	22278	18	63	5326	5290	-	16693	-	34391	-	51084	4000	-	47084
H ₂ O			1906	1885	50	50	2274	2310	-	2491	-	4771	-	7262	-	-	5534
Br			21	20	-	-	-	-	-	14	-	31	-	46	-	-	61
Br ⁻			14	14	-	-	-	-	-	3	-	6	-	9	-	-	32
Co			5	5	-	-	-	-	N/A	4	N/A	9	N/A	13	-	N/A	17
Mn			10	10	-	-	-	-	N/A	8	N/A	17	N/A	25	-	N/A	34
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	10	19	10	28	21	47	-	7	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	4	0	3	0	7	-	0	11
Terephthalic acid			N/A	N/A	-	-	-	-	10764	9	10665	21	21430	30	-	21535	28
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	54	26	40	50	95	76	-	43	11
Benzoic acid			N/A	N/A	-	-	-	-	7	174	6	398	13	572	-	3	679
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	1	0	1	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	17	0	17	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	19	264	0	548	19	812	-	7	1
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	42	0	94	0	136	-	0	9
Bromo toluic acid			N/A	N/A	-	-	-	-	0	30	2	95	2	125	-	0	48
9,10-Dihydro-9,10-dioxo-2,3-Anthracenedicarboxylic acid			N/A	N/A	-	-	-	-	3	0	0	0	3	0	-	5	0
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	1	0
Methane			-	-	120	1	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	9	1	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	184	19	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	0	0	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	0	0	-	-	-	-	-	-	-	-	-	-	-
O ₂	8102	7895	-	-	1253	1253	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	595	619	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	78	112	-	-	-	-	-	-	-	-	-	-	-
N ₂	25997	25334	-	-	25998	25976	-	-	-	-	-	-	-	-	-	-	-
TOTAL	34100	33230	31500	31160	28311	28097	7600	7600	10859	19794	10741	40512	21600	60306	4000	21600	53531

* estimated with Aspen plus^o

25/11/04	0A	0B	1A	1B	2A	2B	3A	3B	4A		4B		5		6	7	
									solid	liquid	solid	liquid	solid	liquid	*	solid	liquid
p-xylene			7109	6991	11	5	N/A	N/A	-	24	-	75	-	99	-	-	24
acetic acid			23059	22676	20	77	5557	5426	-	17008	-	32317	-	49325	4100	-	45225
H ₂ O			1657	1629	46	46	2443	2394	-	2826	-	4942	-	7768	-	-	5232
Br			20	20	-	-	-	-	-	12	-	28	-	39	-	-	55
Br ⁻			14	14	-	-	-	-	-	2	-	5	-	8	-	-	29
Co			5	5	-	-	-	-	N/A	3	N/A	8	N/A	11	-	N/A	16
Mn			10	9	-	-	-	-	N/A	6	N/A	15	N/A	22	-	N/A	30
1,2,4-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	14	31	10	38	23	70	-	10	0
1,2,3-Benzenetricarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
4-(Hydroxymethyl) benzoic acid			N/A	N/A	-	-	-	-	0	3	0	5	1	8	-	0	25
Terephthalic acid			N/A	N/A	-	-	-	-	10837	10	10689	22	21526	32	-	21627	37
4 Carboxy benzaldehyde B A			N/A	N/A	-	-	-	-	49	33	40	77	90	109	-	49	20
Benzoic acid			N/A	N/A	-	-	-	-	7	202	4	424	11	626	-	3	876
Benzophenone-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	1	0	1	-	0	1
o-Toluic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
p-Toluic acid			N/A	N/A	-	-	-	-	34	327	15	722	49	1048	-	10	170
Diphenylmethane-4,4-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	46	0	95	0	140	-	0	72
Bromo toluic acid			N/A	N/A	-	-	-	-	0	37	0	112	0	149	-	0	0
9,10-Dihydro-9,10-dioxo-2,3-			N/A	N/A	-	-	-	-	0	0	0	0	0	1	-	0	1
Anthracene dicarboxylic acid																	
Anthracene-9,10-Dicarboxylic acid			N/A	N/A	-	-	-	-	0	0	0	0	0	0	-	0	0
Methane			-	-	270	1	-	-	-	-	-	-	-	-	-	-	-
Acetaldehyde			-	-	18	1	-	-	-	-	-	-	-	-	-	-	-
Methyl acetate			N/A	N/A	293	23	N/A	N/A	-	-	-	-	-	-	-	-	-
Benzene			-	-	0	0	-	-	-	-	-	-	-	-	-	-	-
Toluene			-	-	1	0	-	-	-	-	-	-	-	-	-	-	-
O ₂	7420.72	7912.08	-	-	683	1148	-	-	-	-	-	-	-	-	-	-	-
CO ₂			-	-	545	541	-	-	-	-	-	-	-	-	-	-	-
CO			-	-	72	100	-	-	-	-	-	-	-	-	-	-	-
N ₂	23811.3	25387.9	-	-	23811	23790	-	-	-	-	-	-	-	-	-	-	-
TOTAL	31232	33300	31860	31330	25770	25731	8000	7820	10941	20569	10759	38880	21700	59449	4100	21700	51783

* estimated with Aspen plus[®]

REFERENCES

- 1) N. S. Allen, M. Edge, J. Daniels, D. Royall, Poly. Degradation and Stability, 62 (1998) 373
- 2) J. E. McIntyre, Chem. Eng. Monogr., 15 (1982) 400
- 3) US 4,782,181 (1988)
- 4) S. H. Jhung, Bull. Korean Chem. Soc., 2002, 23(3), 503
- 5) S. H. Jhung. K. H. Lee, Y. S. Park, Bull. Korean Chem. Soc., 2002, 23(1), 59
- 6) S. H. Jhung. K. H. Lee, Y. S. Park, Applied Catalysis A, 2002, 230, 31
- 7) J. S. Yoo, S. H. Jhung. K. H. Lee, Y. S. Park, Applied Catalysis A, 2002, 223, 239
- 8) US 6 194 607
- 9) WO 03/095408
- 10) W. Partenheimer, Cat. Today, 23 (1995) 69
- 11) PTA Technology Package – Analytical manual.

